

Characteristics of Forces — Mark schemes

PAPERS A TO D • 30 MARKS EACH • TUTOR COPY

Marks in square brackets show how the total is split. Award method marks for a correct formula and a correct substitution even when the arithmetic fails. Almost every mark lost on this topic goes one of four ways: a weight written in kilograms, a resultant given with no direction, opposing forces added instead of subtracted, and a steady speed treated as evidence of a forward resultant. The notes under each answer flag the specific mistake that question was built to catch.

Paper A — Medium

30 MARKS

- 1 A force is a push or a pull on an object (from an interaction with another object) **[1]** [2]
Force is measured in **newtons (N)** **[1]**
Reject "energy" and "power" outright. They are different quantities with different units, and this is the one place on the paper where the definition alone is worth a mark.
- 2 (a) **air resistance** (accept drag) [3]
(b) **weight** (accept gravity / gravitational force)
(c) **tension**
In (b), "gravity" alone is acceptable at this level, but "weight" is the term the syllabus uses and the one to encourage.
- 3 (a) **contact** [3]
(b) **non-contact**
(c) **contact**
Upthrust is the one that gets guessed wrong. The liquid or gas is touching the object, so it is a contact force.
- 4 Change its **speed** (speed it up or slow it down) **[1]** [3]
Change its **direction** **[1]**
Change its **shape** **[1]**
"Make it move" and "make it stop" are the same mark, both being a change of speed. Award one for that pair, not two.
- 5 (a) $5 \times 10 = \mathbf{50\ N}$ [4]
(b) $12 \times 10 = \mathbf{120\ N}$
(c) $W = m \times g = 0.5 \times 10$ **[1] = 5 N [1]**
Any answer given in kilograms scores zero for that part, however good the arithmetic. The unit is the whole point of the section.
- 6 (a) $300 \div 10 = \mathbf{30\ kg}$ [3]
(b) $45 \div 10 = \mathbf{4.5\ kg}$
(c) $7 \div 10 = \mathbf{0.7\ kg}$
Multiplying instead of dividing gives 3000 kg, 450 kg and 70 kg. If all three are ten times too big it is one error repeated, not three — award the method once if it is shown.
- 7 Mass is the amount of matter in an object, measured in **kilograms (kg)** **[1]** [3]
Weight is the force of gravity acting on that object, measured in **newtons (N)** **[1]**
Mass is the same everywhere; weight depends on the gravitational field strength where the object is **[1]**
The third mark is the one usually missing. A comparison of the two units alone is worth two, not three.

- 8** (a) $30 - 20 = \mathbf{10\text{ N to the right}}$ [4]
 (b) $45 - 45 = \mathbf{0\text{ N}}$
 (c) Same direction, so add: $12 + 30$ [1] = **42 N to the left** [1]
(c) is the trap: two forces, so the instinct is to subtract. They point the same way, so they add. Withhold the final mark from any answer with no direction on it.
- 9** (a) **0 N** [1] [3]
 (b) It stays **stationary** (it does not move) [1]
 (c) **Balanced** forces [1]
- 10** It carries on at a **constant speed in a straight line** [1] [2]
 because the forces are balanced, so the resultant force is zero and nothing about its motion changes [1]
"It stops" is the commonest wrong answer and scores nothing. A zero resultant maintains the motion; it does not remove it.

Paper B — Medium

30 MARKS

- 1** Arrow pointing **down**, labelled weight (accept a value in N) [1] [4]
 Arrow pointing **up**, labelled normal contact force / reaction / push from the floor [1]
 The two arrows drawn the **same length** [1]
 because the crate is stationary, so the forces are balanced and the resultant is zero [1]
Arrows drawn as plain lines with no arrowhead lose the first two marks. Worth saying out loud once, because it repeats on every forces paper.
- 2** A force has a **size** (magnitude) [1] [3]
 and a **direction** [1]
 The length of the arrow shows the size and the arrowhead shows the direction [1]
"So you can see it" and "to make the diagram clear" score nothing. The mark scheme wants the word direction.
- 3** (a) **newtons (N)** [3]
 (b) the **spring**
 (c) The stretch (extension) also **doubles**
(c) is a reasoning mark, not a recall one. Accept "the extension doubles"; do not accept "the whole spring doubles in length".
- 4** (a) Same direction, so add: $80 + 50 = \mathbf{130\text{ N to the right}}$ [5]
 (b) $90 - 35 = \mathbf{55\text{ N to the left}}$
 (c) **0 N** (balanced)
 (d) Right: $6 + 4 = 10\text{ N}$. Left: 14 N [1]. $14 - 10 = \mathbf{4\text{ N to the left}}$ [1]
(d) is the question of the paper. Two arrows point right and only one points left, and the single one still wins. Counting arrows is the error it is built to catch.
- 5** (a) **4500 N** [1] [4]
 (b) The speed is steady, so the resultant force is zero and the backward force must equal the driving force [1]
 (c) $5200 - 4500 = \mathbf{700\text{ N forwards}}$ [1]; the forces are now unbalanced, so the lorry **speeds up** [1]
In (a) any answer other than 4500 N means the student has not accepted that steady speed implies balance. Read (b) before deciding how much of the topic has landed.

- 6** Balanced forces have a resultant of zero; unbalanced forces have a resultant that is not zero **[1]** [3]
 Balanced: the object keeps moving at the **same steady speed in a straight line [1]**
 Unbalanced: the object changes speed or changes direction **[1]**
An answer saying balanced forces make a moving object stop is worth one mark at most. This is the single most valuable correction on the paper.
- 7** (a) **Opposite to the direction of motion [1]** [4]
 (b) Any two: walking without slipping, brakes on a car or bicycle, tyres gripping the road, holding a pen, a nail staying in a wall, striking a match **[1] [1]**
 (c) Any one: it wears moving parts down, it wastes energy as heat, it slows machinery, it makes engines need oil **[1]**
- 8** (a) **Friction** (between tyres and road, or in the bearings) **[1]**; **air resistance** (drag) **[1]** [4]
 (b) With no forward force from pedalling, only the backward forces are left **[1]**
 so the resultant force acts backwards, opposite to her motion, which is unbalanced and slows her down **[1]**
"She runs out of energy" is common and scores nothing here. The question asks for a force explanation and the mark scheme wants the word resultant.

Paper C — Hard

30 MARKS

- 1** (a) $W = 250 \times 3.7$ **[1] = 925 N [1]** [5]
 (b) $m = 32 \div 1.6$ **[1] = 20 kg [1]**
 (c) $20 \times 10 = 200$ **N [1]**
(b) has to be done by dividing. Multiplying gives 51.2 kg, which would make the object heavier on the Moon than it started — an absurdity the student can spot without knowing the answer.
- 2** Error 1: weight is a **force**, not a mass — they are different quantities **[1]** [4]
 Error 2: the unit should be **newtons**, not kilograms **[1]**
 $W = 45 \times 10$ **[1]**
 $= 450$ **N [1]**
The two error marks must be genuinely separate points. "It should be in newtons" written twice in different words is one mark.
- 3** (a) $15\,000 - 12\,000$ **[1] = 3000 N upwards [1]** [5]
 (b) $900 - 900 = 0$ **N [1]**
 (c) Vertically the forces are unbalanced, so the balloon **rises, speeding up [1]**;
 horizontally the forces are balanced, so it does not speed up or slow down sideways **[1]**
The point of this question is that vertical and horizontal are settled separately. Any answer that adds all four numbers into one sum has missed it entirely.
- 4** (a) **85 N [1]**, because the box moves at a steady speed, so the resultant is zero and friction must equal the pull **[1]** [4]
 (b) $130 - 85$ **[1] = 45 N in the direction of the pull, so the box speeds up [1]**
In (b) a common answer is 130 N, from assuming friction disappeared when the pull increased. Friction was never removed — only outmatched.
- 5** (a) Left total = $9 + 12 = 21$ **N [1]**; $26 - 21 = 5$ **N to the right [1]** [4]
 (b) **5 N to the left [1] [1]**
"Constant velocity" is doing the work in (b): it means the resultant must be zero, so the new force cancels the 5 N exactly. A student answering 26 N has cancelled the wrong thing.

- 6** (a) $W = 80 \times 10 = \mathbf{800\ N}$ [1] [4]
 (b) $800 - 150$ [1] = **650 N downwards** [1]
 (c) **800 N** [1]
(c) needs no arithmetic at all: constant speed means zero resultant, so air resistance equals weight. If a student calculates here, the idea has not landed.
- 7** There is no air in space, so no air resistance, and nothing is touching it, so no friction [4]
[1]
 With no opposing forces there is no resultant force acting on it [1]
 A zero resultant force means nothing about its motion changes [1]
 A force is needed to **change** motion, not to maintain it, so it keeps moving at the same steady speed [1]
This is the whole topic in one question. Any answer containing "it needs a force to keep going" has the physics backwards, whatever else it says.

Paper D — Hard

30 MARKS

- 1** (a) $W = 40 \times 10 = \mathbf{400\ N}$ [1] [5]
 (b) **400 N upwards** [1], because the crate does not move vertically, so the vertical forces must be balanced [1]
 (c) $260 - 100$ [1] = **160 N in the direction of the pull**, so the crate **speeds up** [1]
The 400 N must not appear anywhere in (c). Dragging the weight into the horizontal sum is the standard error in four-force questions and is worth marking explicitly.
- 2** (a) $190 - 145$ [1] = **45 N to the west** [1] [5]
 (b) The sledge would move west, so friction acts **east** [1]; $45 - 15 = \mathbf{30\ N\ to\ the\ west}$ [1]
 (c) The forces are unbalanced, so the sledge **speeds up towards the west** [1]
Friction is set by the direction of motion, not by the stronger child. An answer putting friction west has stopped thinking about what friction is for.
- 3** (a) $W = 60 \times 10 = \mathbf{600\ N}$ [1] [4]
 (b) **600 N** [1]; the lift moves at a constant speed, so the resultant force on the passenger is zero [1]; so the upward push from the floor must exactly equal the 600 N weight [1]
Moving upwards tempts students into making the upward force bigger. Constant speed rules that out — the direction of travel is irrelevant to the balance.
- 4** (a) $1100 - 750$ [1] = **350 N upwards** [1] [4]
 (b) A resultant force changes motion rather than causing it [1]; the upward resultant is slowing her down, but she was already moving downwards and carries on doing so, just more slowly [1]
This separates students who have accepted that force and direction of travel are independent from those who have not. Expect it to be the least-answered question on the paper.
- 5** (a) $g = W \div m = 138 \div 60$ [1] = **2.3 N/kg** [1] [4]
 (b) $60 \times 10 = \mathbf{600\ N}$ [1]
 (c) **60 kg** [1]
(c) is a one-word answer that a surprising number of students calculate. Mass does not change with location, which is the whole reason the part is there.

- 6** (a) Any two, with directions: **air resistance / friction backwards, weight downwards, normal contact force upwards** [1] [1] [4]
(b) The car travels at a steady speed, so the forces must be balanced [1]; the forward and backward arrows must therefore be drawn the **same length** [1]
Award (a) only where a direction is given. A named force with no direction is half a force arrow and is worth nothing on a diagram question.
- 7** (i) The mistake: weight is a force, so it cannot be measured in kilograms [1] [4]
 $W = 20 \times 10 = \mathbf{200\ N}$ [1]
(ii) The mistake: a steady speed means the forces are **balanced** and the resultant force is **zero** [1]
A resultant force is needed to **change** motion, not to maintain it, so no forward resultant is required [1]
These are the two most common wrong sentences in the whole topic. If a student can correct both here but not on Paper A question 10, the problem is confidence rather than knowledge.